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Proactive Tubing Integrity Assessment Using Predictive Caliper Simulation for Safety, Reliability, and Production Continuity

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Abstract

This paper introduces a novel predictive, non-intrusive methodology for tubing integrity management that combines corrosion analytics with caliper simulation. By integrating real-time data on lift performance, nodal pressure, and fluid properties, the system estimates localized degradation rates and ranks tubing segments based on their failure probability. This approach enables early risk detection, providing operators with data-driven insights for targeted interventions and more efficient use of caliper logging.

The primary outcome of this methodology is optimized intervention planning. By shifting from broad, periodic caliper logging campaigns to exception-based, data-informed monitoring, the frequency of site campaigns is significantly reduced, leading to enhanced operational efficiency and reduced production interruptions. Additionally, this approach supports the extension of tubing service life, while improving safety performance by minimizing the risk of unexpected well failures. The methodology has been successfully trialled by a major operator, with documented case study demonstrating its effectiveness and providing a robust validation of its predictive capabilities.

The methods employed include the collection of real-time pressure and fluid data, as well as production parameters, to feed into a mechanistic corrosion-erosion model. This model simulates wall loss progression and generates virtual caliper profiles to predict the internal condition of the tubing. The ranking system, which scores tubing segments based on their failure risk, allows for focused surveillance and resource allocation. In turn, this optimization reduces the need for broad caliper logging efforts, resulting in considerable cost savings.

In the results and observations section, it is demonstrated that the ranking system accurately identifies vulnerable tubulars, allowing for proactive intervention before catastrophic failures occur. The non-intrusive prediction method reduces the frequency of interventions and production interruptions, while continuous monitoring ensures early detection and supports better decision-making. The use of predictive modelling has shown to extend tubing service life and minimize late-stage integrity failures. Furthermore, a successful field trial validated the approach, which has been fully deployed with real-world case studies.

This paper introduces a digital-first approach to well integrity management, blending corrosion forecasting with virtual caliper simulation. The predictive ranking system enables focused inspections, optimizing resource use while improving safety and reliability.

Introduction

Tubing is a critical component of oil and gas wells, acting as the primary barrier that ensures the well's integrity by containing pressure and preventing fluid migration. This is particularly important in wells that utilize packers, where the tubing needs to maintain pressure containment throughout the life of the well. It is also responsible for ensuring the safe and efficient flow of fluids from the reservoir to the surface. Any compromise in the tubing's integrity can lead to catastrophic events, including blowouts, loss of well control, and significant production losses. Tubing failures not only disrupt production but also introduce substantial safety risks and financial implications. The oil and gas industry operates under strict safety regulations, with well integrity being a focal point of operational safety management.

The primary function of tubing in a well is to serve as the conduit for the extracted fluids, such as oil, gas, and water, while preventing unwanted fluid migration between different zones within the wellbore. This makes it an essential barrier to ensure both the safety and productivity of the well. Over time, however, the tubing faces degradation from several factors, including corrosion, erosion, mechanical wear, and fluid dynamics. Corrosion, driven by chemical interactions between the tubing material and corrosive elements such as hydrogen sulphide (H_2S), carbon dioxide (CO_2), and water, is one of the most significant contributors to tubing degradation. Erosion, which is caused by the high-velocity flow of particulates, can also lead to material loss and damage, compromising the tubing's ability to withstand internal pressures and ensuring safe fluid containment.

However, these degradation mechanisms are not always immediately visible. Tubing degradation often progresses slowly, and the first signs of failure may not be detected until it is too late to prevent major damage. By the time pressure anomalies are identified, indicating fluid migration, the tubing may have already been severely compromised by corrosion or erosion, leading to operational shutdowns, extensive repairs, and potentially, catastrophic well failures. This makes early detection and proactive management of tubing integrity a high priority for oil and gas operators.

Corrosion is an insidious process that often goes unnoticed until it has caused significant damage. The presence of corrosive gases such as CO_2 and H_2S in production fluids increases the risk of tubing failure. These gases, when combined with the temperature and pressure conditions found within the well, promote corrosion by creating an electrochemical reaction on the tubing material. Similarly, the erosion process accelerates in wells with high flow rates and the presence of particulate matter, such as sand or scale, which physically wears away the tubing material. While erosion is generally more localized to specific regions of the well, corrosion tends to be more widespread, affecting the entire length of the tubing over time.

Both of these degradation mechanisms can result in wall thinning, pitting, and eventual tubing rupture if left unaddressed. Furthermore, tubing failure is not just an issue of structural integrity; it also impacts production. In many cases, corroded or eroded tubing can reduce flow efficiency, resulting in lower production rates. Moreover, if left untreated, corrosion and erosion can lead to more severe integrity issues, including catastrophic failures such as blowouts or the loss of well control.

The challenge for operators is that corrosion and erosion are often not visible until significant damage has already occurred. Wells that are experiencing corrosion or erosion issues may still be producing fluids, but at reduced efficiency or with heightened safety risks. As a result, operators must find a way to assess the integrity of the tubing continuously, to ensure that any degradation is detected before it leads to a complete failure.

Traditionally, well integrity has been monitored through periodic methods, such as caliper logging, pressure testing, and visual inspections. These methods remain valuable tools for operators but have significant limitations when it comes to providing real-time data and early warnings of tubing degradation.

Caliper logging is one of the most widely used methods for assessing tubing integrity, where a tool is lowered into the wellbore to measure the internal diameter of the tubing. This technique helps operators identify wall thinning and other forms of corrosion. However, despite its widespread use, caliper logging has notable limitations. First, it is intrusive, as it requires well shutdowns to insert and remove the tool. Second, it offers only periodic snapshots of the tubing's condition and often fails to detect degradation that occurs between logging intervals. As a result, significant tubing degradation may go undetected until the next scheduled caliper survey, potentially leading to unplanned shutdowns and expensive repairs.

Pressure testing is another method used to evaluate tubing integrity by subjecting the tubing to high pressures and checking for leaks. However, this method is also intrusive and typically only provides information about the overall structural integrity of the tubing. It does not provide specific details about localized degradation or the risk of future failure.

Visual inspections can help identify surface-level issues such as leaks or external corrosion. However, these inspections only provide a limited view of the tubing's condition and fail to identify issues below the surface or in areas that are not easily visible.

Each of these methods is valuable but offers only limited information about the tubing's overall health. Furthermore, they are reactive, often identifying issues only after they have already affected the well. Given the high cost of running caliper surveys on multiple wells, this reactive approach is inefficient and costly, particularly in fields with a large number of wells.

In fields with a significant number of wells, the costs of running regular caliper logs can be prohibitive. While caliper logging is an effective tool for detecting tubing degradation, it requires a significant investment of time and resources. Each caliper survey requires the deployment of logging tools, which must be inserted and retrieved from the wellbore, often requiring the well to be shut in during the process. This downtime not only reduces production but also introduces additional costs for mobilizing equipment, labour, and specialized personnel.

In fields with hundreds of wells, it becomes increasingly difficult to justify the cost of performing caliper surveys on every well. Operators must prioritize wells based on perceived risk, often relying on the assumption that only a subset of wells may be experiencing integrity issues. This assumption, however, is not always accurate, and it can lead to underestimation of the true scope of degradation across the field. As a result, wells that may be experiencing early signs of corrosion or erosion could go undetected, leading to more severe failures down the line.

Moreover, the financial burden of performing caliper logging across an entire field can limit the ability of operators to monitor wells at the frequency required to proactively manage integrity. Consequently, tubing failures may occur unexpectedly, leading to unplanned production downtime, costly repairs, and, in some cases, catastrophic well failures.

Introducing a Predictive Approach to Tubing Integrity Management

To address these challenges, this paper introduces a predictive, non-intrusive methodology for tubing integrity management that integrates real-time operational data, fluid properties, and mechanistic models for corrosion and erosion prediction. This system leverages predictive simulations, allowing operators to assess the integrity of the tubing.

By integrating data from real-time pressure readings, fluid analysis, and production parameters, the system can simulate the corrosion and erosion processes occurring within the wellbore. Using mechanistic models, the system predicts localized degradation rates and ranks tubing segments based on their failure risk. This enables operators to identify areas of concern and schedule maintenance or interventions proactively, reducing the risk of unexpected failures.

This predictive methodology offers operators a continuous overview of tubing integrity, reducing the need for extensive, widespread caliper surveys across hundreds of wells in a field. This method helps to optimize caliper log run interventions schedules, improve decision-making, and reduce downtime. By identifying potential failures before they occur, the approach not only reduces operational costs but also enhances safety by minimizing risks to personnel, equipment, and the environment.

This methodology significantly reduces the need for broad-scale, field-based caliper surveys, which are both costly and time-consuming. Instead, the system provides a data-driven, predictive approach that can be deployed across an entire field, offering insights into the status of all wells without the need for physical intervention. This allows operators to optimize maintenance planning, allocate resources more effectively, and prioritize wells at the greatest risk of failure.

Furthermore, this approach contributes to improved operational efficiency and safety by providing continuous real-time data on tubing integrity. The system allows operators to track well conditions continuously, reducing the risk of unplanned downtime and enabling more effective long-term asset management.

The challenges of tubing corrosion and the costs associated with traditional methods of tubing integrity monitoring are significant. As the oil and gas industry faces increasing cost constraints and demands for operational efficiency, new approaches to well integrity management are essential. By integrating real-time operational data with mechanistic corrosion and erosion models, predictive simulations offer a more effective and cost-efficient way to manage well integrity, delivering significant cost savings, enhanced safety, and extended asset life.

Background and Literature Review

Current Techniques for Tubing Integrity Monitoring

Caliper logging and pressure testing are two of the most widely used traditional methods for assessing tubing integrity in oil and gas wells. These methods rely on periodic inspections to monitor the condition of the tubing. In caliper logging, a specialized tool is run into the wellbore to measure the internal diameter of the tubing. By doing so, it can detect wall loss, often resulting from corrosion, by identifying any increase in the tubing's inside diameter. Pressure testing, on the other hand, involves subjecting the tubing to internal pressures to verify its ability to contain fluids without leaking, helping to ensure that the well can withstand operational pressures without failure.

However, both methods are inherently reactive and have significant limitations. Since they are typically performed only when there are signs of tubing failure or degradation—such as pressure anomalies, leaks, or other operational issues—they often detect damage only after it has already occurred. As a result, by the time the issue is identified, the tubing may already be severely degraded, requiring costly repairs or even full replacement. This delay in detection means that proactive maintenance is not possible, and unexpected failures can still occur, leading to unplanned downtime, increased operational risks, and loss of production.

Moreover, both caliper logging and pressure testing are intrusive methods. They require the well to be shut in during the procedure, resulting in operational downtime. This shutdown time not only disrupts production but also incurs additional costs related to equipment mobilization, personnel, and other logistical requirements. For fields with a large number of wells, performing such inspections across all wells can be prohibitively expensive and logistically challenging, reducing the overall efficiency of well integrity management.

Given these limitations, there is an increasing need for continuous, non-intrusive monitoring systems that can provide real-time data on tubing integrity. These systems would allow operators to monitor the predicted condition of the tubing continuously, predicting early signs of degradation before they evolve into major issues. By enabling proactive well management, operators would be able to make informed decisions on when and where to intervene, preventing unexpected failures and minimizing downtime. Such predictive

systems would significantly reduce operational costs, enhance safety, and improve overall well performance by providing timely, accurate data on the well's condition with reduced need for intrusive inspections.

Corrosion and Erosion Mechanisms in Tubing

Corrosion and erosion are the primary mechanisms that degrade tubing over time. Corrosion is caused by electrochemical reactions between the tubing material and aggressive well fluids, such as hydrogen sulphide (H_2S), carbon dioxide (CO_2), and water. The rate of corrosion can be modelled based on various factors, including fluid composition, temperature, pressure, and flow velocity.

Erosion, on the other hand, occurs due to the mechanical wear caused by high-velocity flow of particulates, such as sand or scale, through the tubing. The erosion rate is influenced by factors such as flow velocity, particle size, and tubing material hardness.

These degradation processes not only compromise the structural integrity of the tubing but also lead to increased operational risks, such as leaks, blowouts, and failure of the wellhead system.

Predictive Modelling in Well Integrity Management

Predictive modelling has dramatically transformed well integrity management by providing powerful tools to simulate degradation processes within tubing and forecast the future condition of wells. These sophisticated models leverage a wide array of real-time data, including pressure, temperature, fluid properties, and production parameters, to generate accurate, dynamic predictions of tubing performance over time. The integration of such real-time data ensures that the models reflect current well conditions, allowing for more precise forecasting of tubing degradation and operational risks. The application of mechanistic corrosion and erosion models further enhances this process by simulating the physical processes that cause tubing degradation, such as material loss from corrosion and mechanical wear from erosion. This level of detail provides operators with a deeper, more comprehensive understanding of tubing health, enabling them to take proactive steps in addressing potential failures before they occur, ultimately leading to safer, more efficient, and cost-effective well management.

Methodology

Multiphase Flow Nodal Analysis Model

To effectively predict the corrosion and erosion impacts on tubing, the first step is building a multiphase flow nodal analysis model of the well. This model divides the well into multiple segments or "nodes," each representing a specific point in the wellbore. The fluid properties, such as pressure, temperature, and fluid velocities, are predicted at each node. The following parameters are calculated at each node:

- **Pressures:** The pressure at each node is derived from the flow dynamics and operational conditions of the well, accounting for factors such as production rates and wellbore geometry.
- **Temperatures:** The temperature at each node is determined by considering the geothermal gradient and heat transfer between the fluids and surrounding rock formations.
- **Fluid Velocities:** The velocity of fluids in the wellbore is modelled, incorporating the effects of friction, flow rate, and wellbore dimensions.
- **Produced Fluid Properties:** The properties of the produced fluids, including density, viscosity, and chemical composition, are predicted at each node, considering how the fluid interacts with the wellbore and surrounding environment.

This model is updated at critical time steps in the well's life to capture changes in the operational conditions and fluid properties. It provides a dynamic view of the well's performance and allows for accurate corrosion and erosion predictions.

Corrosion and Erosion Models

Corrosion Models. Several well-established models are commonly used in the oil and gas industry to predict corrosion rates in tubing, each offering unique methodologies based on different corrosion mechanisms and environmental conditions:

- **De Waard and Milliams Model:** One of the most widely utilized models, it predicts corrosion rates based on H₂S concentration and partial pressure. Minor discrepancies can arise under transient operational conditions. This model is considered a standard for wells with high H₂S concentrations (De Waard, Lotz & Milliams, 1991; De Waard & Lotz, 1993; De Waard, Lotz & Dugstad, 1995).
- **Cassandra 98 Model:** A BP-modified version of the De Waard and Milliams model, incorporating operational experience to reflect BP-specific conditions (McMahon & Paisley, 1997).
- **NORSOK M-506 Standard:** Provides corrosion prediction and management guidelines, accounting for temperature, pressure, fluid composition, and formation of protective films (NORSOK, 2017).
- **Oldfield, Swales, and Todd Model:** A first-pass model for oxygen corrosion in carbon steel. It tends to overpredict corrosion rates but is useful for initial screening (Oldfield, Swales & Todd, 1981).

Erosion Models

Erosion is another critical factor affecting tubing integrity, particularly in wells with high-flow rates or those that produce solids, such as sand. Erosion models help predict the mechanical wear that can result in the thinning or failure of tubing. The key models for erosion prediction include:

Erosion is critical in wells with high-flow rates or solids production. Key predictive models include:

- **Tulsa and Jordan Models:** The Tulsa model, developed at the University of Tulsa, predicts erosion in piping systems and is incorporated in tools like the Sand Production Pipe Saver (SPPS). The Jordan model refines these predictions with additional flow and operational factors (Jordan, 1998; Martin, 1999).
- **API RP14E Guidelines:** Provides a framework for design and analysis of subsea and surface facilities in erosive service, widely adopted in the oil and gas industry (Shirazi et al., 1994; McLaury & Shirazi, 1999).
- **Salama and Venkatesh Model:** This model is used for estimating erosion in high-velocity flow conditions where solid particles are present in the production stream. It is an enhancement of earlier models and considers factors such as particle size, velocity, and material hardness to predict erosion rates more accurately (Salama & Venkatesh, 1996).

Scaling and Deposition Models

Scale formation in tubing can reduce flow efficiency and damage equipment. Several models predict the risk and rate of scale deposition:

- **Calcium Carbonate and Sulfate Scaling Models:** Predict scale formation using water chemistry, temperature, and pressure conditions. Models often rely on empirical correlations validated against field measurements (McCain, 1990; Standing, 1977).
- **Supersaturation and Solubility Models:** Determine the onset of scale precipitation based on solubility limits and supersaturation indices, providing guidance for scale inhibition treatments (Tchobanoglous et al., 2014).
- **Corrosion–Scale Interaction Models:** Some models integrate corrosion and scale formation effects, particularly where scale layers can inhibit or accelerate localized corrosion. These are often

calibrated using field data and predictive simulations (De Waard & Lotz, 1993; Kermani & Harrop, 2019).

Application and Validation

All models described are widely used in well integrity and production optimization. Comparisons with historical field data or lab measurements are essential to validate model predictions, ensuring that results are robust even under transient operational conditions. Where proprietary modifications exist (e.g., Cassandra 98), validation against internal datasets provides confidence for operational application.

Z-Factor Models and Thermodynamic Corrections

Z-factors, or compressibility factors, are essential for accurately characterizing gas behaviour under varying temperature and pressure conditions. The Z-factor allows conversion between real gas and ideal gas properties, which is critical for flow modelling, reservoir calculations, and multiphase simulations. Several models are commonly used to determine Z-factors:

- Dranchuk and Abou-Kassem Model: This model is widely adopted in the petroleum industry due to its high accuracy over a broad range of gas compositions and pressures. It is particularly effective for high-pressure gas systems, providing reliable compressibility factor calculations for real gas mixtures (Dranchuk & Abou-Kassem, 1975).
- Hall and Yarborough Model: Although less commonly used than the Dranchuk model, the Hall and Yarborough correlation provides a simpler empirical approach to Z-factor calculation. It is often applied where computational simplicity is preferred, and the gas composition is near-ideal (Hall & Yarborough, 1973).
- Thermodynamic Corrections: In some cases, additional thermodynamic corrections are applied to account for non-ideal behaviour at extreme pressures and temperatures, particularly for high CO₂, H₂S, or heavy hydrocarbon content in the gas (Standing, 1977; McCain, 1990).

Corrosion Prediction Model

Once the multiphase flow model is established, the next step is to predict corrosion rates at each node. The corrosion rate, denoted as R_{corr} , is calculated using established models such as De Waard & Milliams, NORSOK, and others. These models consider multiple factors including temperature, pressure, and fluid composition.

Corrosion and Erosion Mechanisms in Tubing. Corrosion in tubing is a result of electrochemical reactions between the tubing material (typically carbon steel) and aggressive well fluids, including water, gas, and crude oil containing corrosive agents such as H₂S or CO₂. The corrosion R_{corr} can be expressed as:

$$R_{corr} = k \cdot C_{H_2S} \cdot P_{H_2S}^n$$

Where:

$R_{(corr)}$ is the corrosion rate (mm/year),

k is a constant,

C_{H_2S} is the concentration of hydrogen sulphide

P_{H_2S} is the partial pressure of hydrogen sulphide,

n is an empirical constant

Additional factors considered by the model include:

- Temperature and pressure.
- Water cut and oxygen content (especially in injection wells).

- Protective film formation (e.g., FeCO_3).
- Erosion-corrosion interactions.
- Hydrogen sulphide (H_2S) corrosion mechanisms.

Erosion, on the other hand, is caused by the physical impact of high-velocity flow of particulates (e.g., sand, scale) against the tubing walls, leading to material wear. The erosion rate R_{erosion} is influenced by flow velocity v , particle size d , and material hardness H and is given by the following equation (Levens et al., 2020):

$$R_{\text{erosion}} = \frac{\alpha \cdot v^3 \cdot d^2}{H}$$

α is an empirical constant

v is the flow velocity

d is the particle size

H is the hardness of the tubing material

Erosion is particularly critical in systems where high-flow velocities and particulates (such as sand and scale) are present in the production fluids.

Virtual Caliper Simulation

The virtual caliper simulation provides predictive profiles of tubing integrity by integrating real-time data on corrosion and erosion degradation rates into a model of the tubing geometry. This process simulates the internal conditions of the tubing and continuously updates the tubing condition profiles as new data is received. The simulation predicts the future condition of the tubing, including changes in wall thickness due to corrosion and erosion, offering a dynamic and accurate representation of both current and projected tubing states.

The virtual caliper model simulates the structural integrity of the tubing by predicting wall loss progression over time. The model uses the following equation to calculate the change in diameter due to degradation:

$$\Delta D = \int_{t_0}^t R_{\text{total}}(t) dt$$

ΔD is the change in diameter due to wall loss,

$R_{\text{total}}(t)$ is the total degradation rate as a function of time.

This equation integrates the degradation rate over time, factoring in elements such as corrosion and erosion to provide an accurate representation of the tubing's condition at any given moment. Unlike traditional caliper surveys, which offer only periodic snapshots of the tubing's internal diameter, this predictive system simulates degradation progression in real-time. The model dynamically incorporates variables such as pressure, temperature, flow velocities, and fluid composition, predicting how these factors influence corrosion and erosion over time. Historical caliper survey data is also used as a "tuning factor" to refine the final prediction, enhancing the robustness of the forecast.

The nodal analysis model integrates multiple inputs to predict well integrity degradation, including real-time wellhead pressure (WHP), temperature, flow rates, fluid properties, vertical lift performance, and historical caliper survey data. Where available, additional operational measurements, such as permanent downhole gauge (PDHG) data, are incorporated to further refine corrosion and erosion forecasts. This combination of operational and historical data enables the model to simulate future degradation accurately, accounting for both fluid dynamics and well-specific geometrical factors.

This approach allows operators to assess tubing integrity without intrusive interventions, such as traditional caliper logging, which typically requires well shutdowns. The system continuously updates by

capturing changes in operational conditions and fluid properties at critical intervals throughout the well's life. This ensures robust forecasting of degradation processes, providing operators with the information needed to make proactive and informed decisions regarding maintenance planning, targeted caliper surveys, and intervention strategies.

Ranking System for Tubular Segments

Each segment of tubing is assigned a failure risk score based on its degradation rate and the criticality of its role in maintaining well integrity. The failure risk R_{risk} for a given segment is computed using the following equation:

$$R_{risk} = \frac{R_{total}}{L_{life}}$$

Where:

- R_{risk} is the failure risk,
- $R_{(total)}$ is the total degradation rate for the segment, considering both corrosion and erosion processes,
- L_{life} is the estimated remaining life of the segment.

Segments with high-risk scores are flagged for immediate attention, while lower-risk segments are scheduled for monitoring at later intervals. This ranking system allows operators to focus resources on high-priority wells and tubing segments, ensuring that interventions are applied where they are most needed and reducing the need for broad, periodic caliper surveys.

Risk-Based Management Approach

A key advantage of the predictive simulation approach is its ability to implement a risk-based management strategy for well integrity. Traditional methods, such as periodic caliper surveys and pressure testing, are typically reactive and offer limited insights into well condition. These approaches identify issues only after they have already impacted the well, failing to capture the dynamic nature of degradation processes.

In contrast, predictive models provide continuous, real-time updates on the well's condition by integrating historical and real-time data, including pressure, temperature, fluid properties, and flow rates. The use of historical caliper survey data further enhances the model by "tuning" the predictions, ensuring more accurate forecasts of future corrosion and erosion conditions. By simulating the effects of variables like fluid composition, flow dynamics, and temperature, the system predicts which areas of the well are most likely to experience significant degradation.

This ability to predict degradation enables operators to focus on the highest-risk wells or tubing segments, optimizing maintenance and interventions. Rather than relying on fixed schedules or inspecting all wells, operators can prioritize wells with the greatest likelihood of failure, ensuring resources are efficiently allocated and minimizing downtime.

The predictive approach enhances both safety and efficiency. Early detection of high-risk areas allows for timely interventions, preventing well failures and reducing the risk of safety hazards, such as pressure changes or blowouts. Proactive maintenance not only protects personnel but also mitigates environmental risks.

Moreover, the use of predictive simulations enables better long-term asset management. Operators can develop data-driven maintenance plans, maximize resource use while minimize unnecessary costs. This approach not only improves cost-efficiency but also ensures sustainable well operations.

Ultimately, adopting a predictive, risk-based management strategy marks a significant shift from traditional reactive methods. It allows operators to better manage well integrity, improving both safety and operational outcomes over time.

Case Study: Field Trial of Predictive Tubing Integrity System for OMV Romania Assets

Trial Setup

A field trial was conducted with OMV Romania to evaluate the effectiveness of a predictive tubing integrity system. The main goal was to simulate and forecast corrosion and erosion rates, allowing the operator to prioritize wells for maintenance or intervention. Real-time operational data, such as pressure, temperature, flow rates, and fluid properties, was gathered from the wells and integrated into the system.

The predictive system utilized a multiphase flow nodal analysis model to forecast future corrosion and erosion conditions for each well, incorporating factors such as vertical lift performance, nodal pressure profiles, and historical caliper survey data to improve the accuracy of degradation predictions.

Results

The predictive system forecasted corrosion and erosion rates across all wells in the field. High-risk wells, identified by predicted corrosion rates, were flagged for immediate intervention, while lower-risk wells were scheduled for monitoring at later intervals.

These simulation results were compared with actual findings from subsequent field-based caliper surveys, showing strong alignment and validating the system's predictive capabilities. Wall thickness forecasts, based on corrosion rate calculations, were provided, enabling proactive management of well integrity and reducing the risk of unexpected failures.

Figure 1 presents the calculated corrosion and erosion profiles along the depth of a representative well. The x-axis represents the predicted wall thickness loss (mm), while the y-axis shows the measured depth (m). The chart overlays multiple data sets, including predicted values from the DeWaard/Tulsa corrosion/erosion model (represented by the orange curve) and measured caliper survey data (represented by the blue curve), enabling a direct comparison and validation of the predictive model.

This visualization provides an intuitive understanding of localized degradation along the tubing, highlighting critical sections where wall loss is most significant. For example, a tooltip on the chart indicates that at a depth of approximately 2,537 meters, the remaining wall thickness two years ago was 7.576 mm, as predicted by the DeWaard/Tulsa model. The blue curve's distinct profile relative to the orange curve represents actual survey data used to validate the model's accuracy.

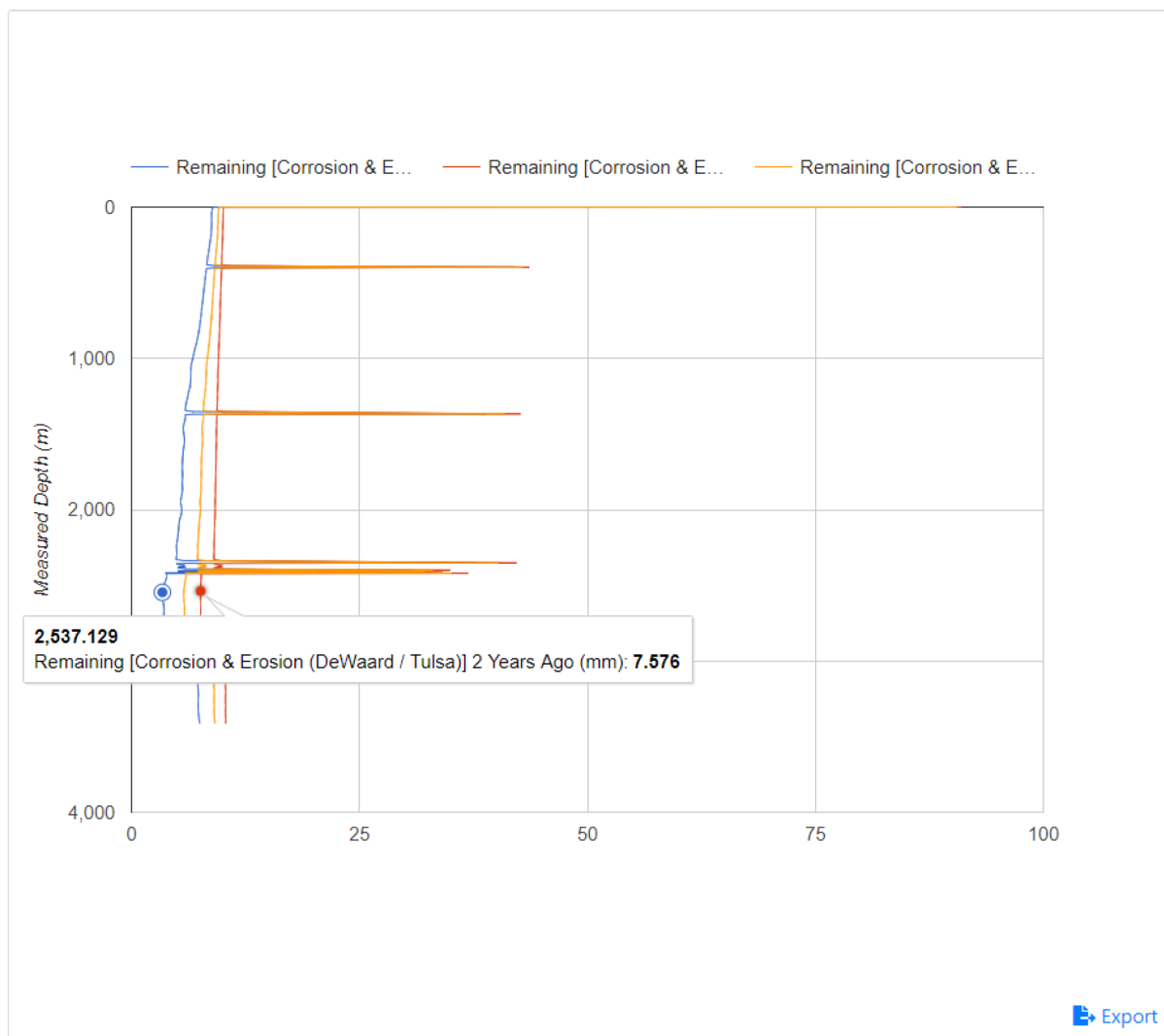


Figure 1—Predicted and Measured Corrosion & Erosion Profiles Along Well Depth

Visualization of Results

Calculations were intuitively represented on a well depth chart, with the ability to overlay calculated parameters against measured values, allowing for easy comparison. The next step in the system involved a 3D representation of the remaining wall thickness over time, which could be cycled through different intervals such as one week, one month, one year, or even a decade. This time-projection feature allowed the operator to understand the predicted failure moment and the depth at which failure was likely to occur. Changes to measured well parameters or projected future conditions were visualized, highlighting critical points along the tubing length. This provided valuable insights into potential failure zones (Figure 2) below.

Figure 2 is a **3D visualization** of the predicted remaining wall thickness along a well. The chart, based on a caliper survey from January 30, 2020, shows the wellbore's structural integrity over a specific length. The model uses the survey data as a baseline and then applies **corrosion and erosion rates** to project how the remaining wall thickness will change over time.

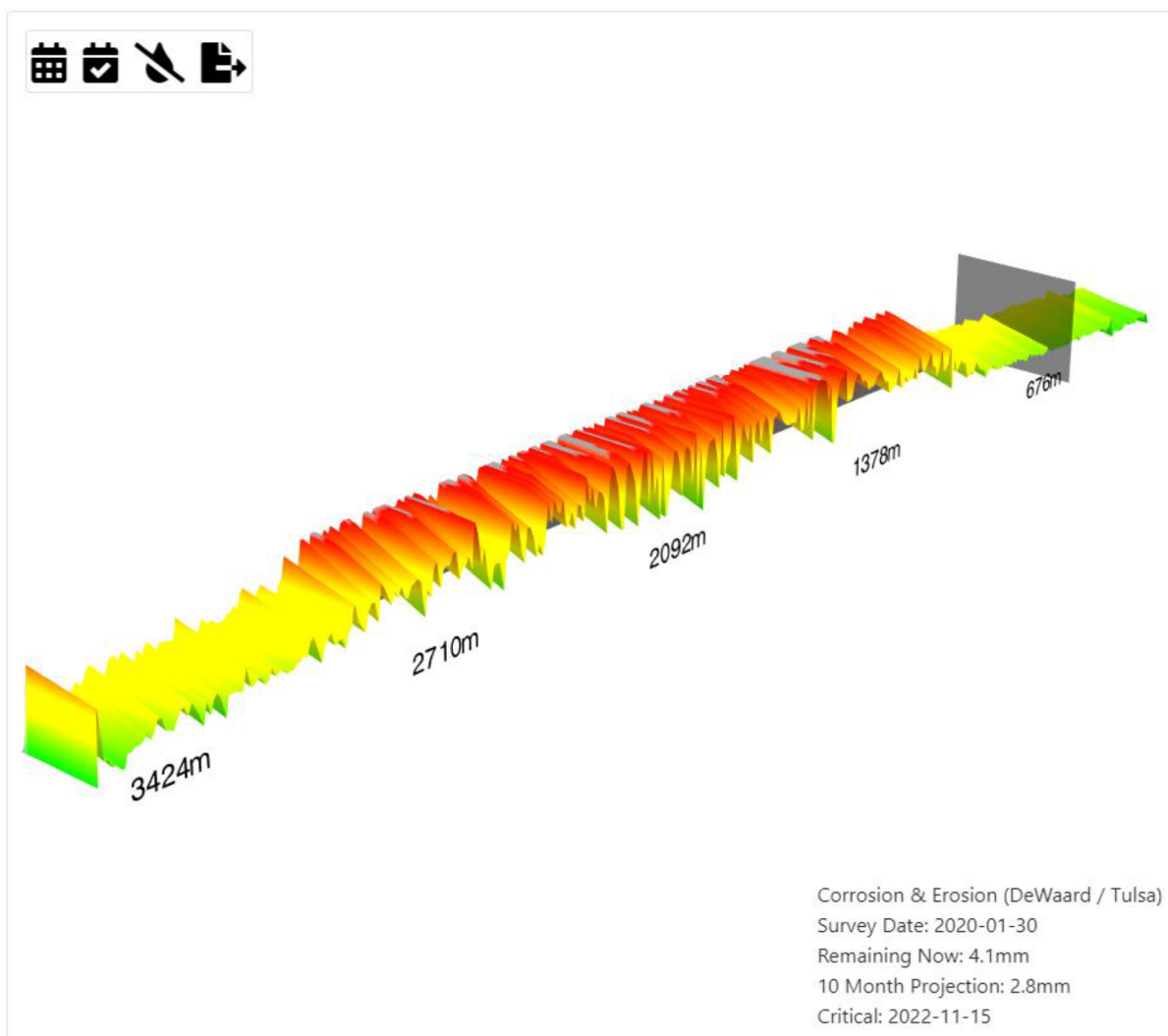


Figure 2—3D visualization of time projection of wall thickness degradation

Key Features

- **Color-Coded "Hot Spots":** The chart uses a colour scale to highlight areas of increased degradation. The red and yellow sections indicate areas with accelerated or uneven wall thickness loss, often caused by factors like changes in flow velocity or fluid composition. These are the **"hot spots"** that require close monitoring.
- **Data Projection:** Based on the baseline survey data, the chart projects future conditions. The text in the bottom-right corner shows the **current remaining wall thickness is 4.1 mm** and projects that it will be **2.8 mm** in 10 months. This projection helps predict when the well might reach a critical state, which is noted as November 15, 2022.
- **Baseline Data:** If actual caliper survey data isn't available, the system can still generate this visualization by using estimated corrosion and erosion rates from the initial equipment installation date to calculate a baseline. This ensures the model can provide useful insights even with limited initial data.

Impact on Operations and Value

The predictive tubing integrity system brought significant value to the operator's well integrity management process. Detailed, scheduled reports were generated, providing key tubing corrosion parameters and

enabling easy visualization of the entire well stock. This approach gave the operator a clearer understanding of tubing remaining life and identified critical parameters affecting well integrity.

These reports were further filtered to generate prioritized lists for tubing replacement, which was a key goal of the initiative. By focusing resources on the highest-risk wells, the operator was able to optimize maintenance schedules and reduce unnecessary interventions, leading to significant cost savings.

An additional benefit was the continuous improvement of the corrosion model calculations. The model evolved over time based on actual well integrity data, making it applicable even to wells without direct corrosion measurements. This ensured that all wells, regardless of their data availability, benefited from the predictive system's capabilities.

This innovative approach enhanced the operator's ability to make informed decisions regarding well integrity and tubing maintenance, improving operational efficiency and reducing the reliance on broad-scale, intrusive interventions like field-based caliper surveys. Ultimately, the trial demonstrated how predictive modelling can be integrated into well integrity management systems, optimizing well safety, reliability, and production continuity.

Conclusion

The introduction of predictive simulation tools for tubing integrity management marks a significant advancement in well integrity monitoring. By integrating real-time data, previous log evaluation data, mechanistic models, and advanced corrosion prediction techniques, this solution enables operators to proactively manage well integrity, reduce operational costs, and optimize intervention planning.

The field trial demonstrated the effectiveness of the system in predicting future corrosion conditions, highlighting its potential to complement traditional caliper surveys by enabling more targeted, cost-effective monitoring. Traditional methods, such as physical caliper logging, typically require downtime and are reactive, providing only snapshots of tubing condition. In contrast, this predictive system offers continuous, real-time monitoring, allowing operators to detect and address potential issues before they develop into significant failures. This proactive approach improves decision-making regarding maintenance schedules, resource allocation, and intervention planning, ultimately leading to enhanced operational efficiency.

As the oil and gas industry continues to face budget constraints and operational challenges, such predictive tools are becoming essential. The ability to forecast and manage tubing degradation in a proactive, data-driven manner provides operators with a more sustainable approach to well integrity management. This solution not only reduces the costs associated with traditional inspection methods but also ensures that well performance and safety are maintained effectively throughout the life of the well. By improving operational efficiency, safety, and the long-term performance of wells, this predictive methodology represents a critical innovation in the ongoing efforts to optimize well integrity management in the industry.

Future Work and Enhancements

As the oil and gas industry continues to push for more efficient, cost-effective, and safer operations, further development and enhancements to existing corrosion and erosion prediction models are essential. While the models discussed above provide significant value in predicting corrosion and erosion rates based on regional data, there remains room for improvement and expansion to enhance their applicability and accuracy. Here are several key areas for future work and potential enhancements:

Expansion of Regional Empirical Models

Although the current models provide critical insights into specific regional conditions, further development is needed to incorporate a broader range of empirical correlations. This will enable more accurate predictions across different geographical regions, particularly in emerging markets and areas where less data is available. For example, regional models could be expanded to account for additional environmental factors,

such as varying microbial populations or unique fluid compositions encountered in regions not currently covered by existing models. Additionally, integrating more localized field data into these models will improve the predictions and make them more adaptable to changing well conditions.

Integration of Machine Learning and Artificial Intelligence (AI)

The integration of machine learning (ML) and AI into corrosion and erosion models offers significant potential. These technologies can analyse large volumes of operational data, uncovering patterns that traditional methods may miss. ML allows predictive models to evolve continuously based on new data, improving the accuracy of predictions over time. However, for the models to be effective, they need to learn from diverse datasets. Using data from just one client's wells is insufficient. A broader range of operational data is required to ensure the model's accuracy and reliability, ultimately providing more actionable insights for well integrity management.

Enhanced Predictive Analytics for Erosion and Scaling

While the current models for erosion and scaling provide valuable insights, there is room to further enhance these models by incorporating more granular data on flow dynamics, particulate behaviour, and scaling tendencies. For example, erosion is highly influenced by factors such as flow rate, particle size, and material hardness, which can vary significantly depending on operational conditions.

Integration of Microbial Influence on Corrosion

The role of microbial-induced corrosion (MIC) in many regions, particularly in tropical and humid environments like Colombia, is significant and warrants further exploration. While some models, such as the Farshad, LeBlanc, Garber, and Osorio model, account for microbial activity, further enhancements could be made by incorporating more detailed data on microbial populations and their effects on corrosion mechanisms.

Real-Time Monitoring and Digital Twin Integration

Incorporating real-time monitoring technologies and integrating digital twin technology into the corrosion and erosion models will significantly enhance the ability to predict and manage well integrity. A digital twin is a virtual representation of a physical asset, updated in real-time with data from sensors deployed within the well or the tubing.

Cross-Model Integration for Comprehensive Well Integrity Management

Finally, an area for future work is the integration of multiple models—corrosion, erosion, scaling, and microbial activity—into a unified, comprehensive well integrity management system. Currently, these models tend to operate in isolation but combining them into a single framework that considers all degradation factors simultaneously will provide a more holistic view of tubing integrity.

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